Вероятностное машинное обучение: введение

Автор: Кевин П. Мёрфи

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Вероятностное машинное обучение: введение — принципы, алгоритмы, применение и инженерная практика

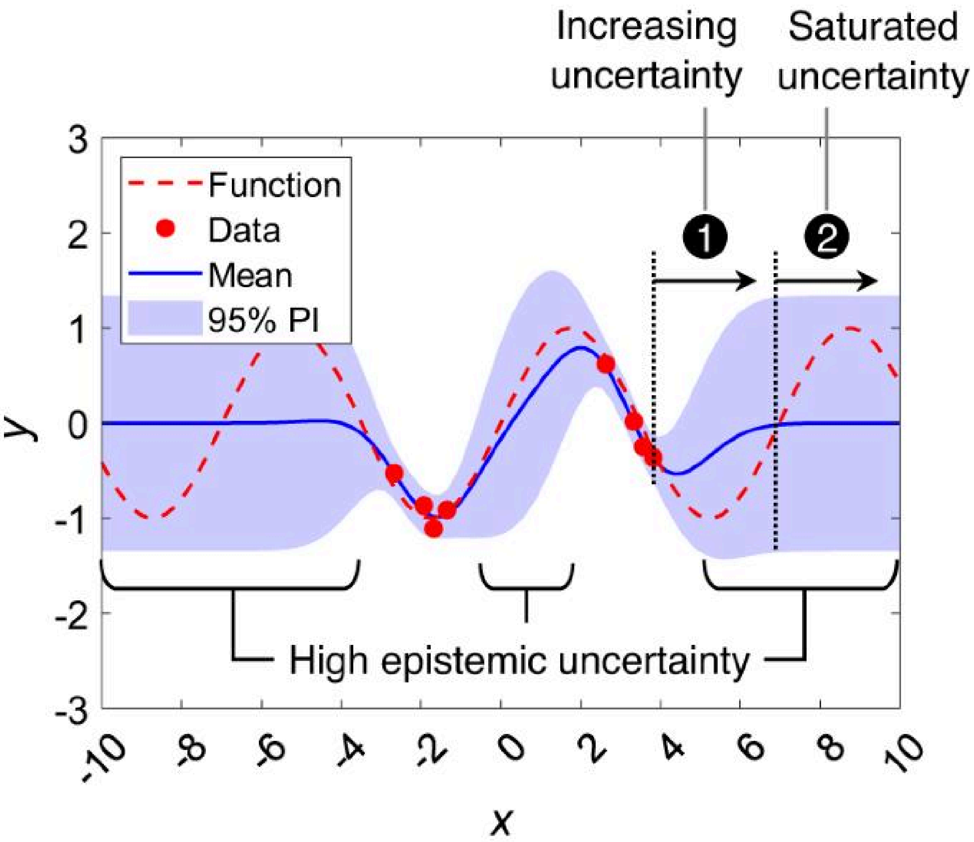
Введение

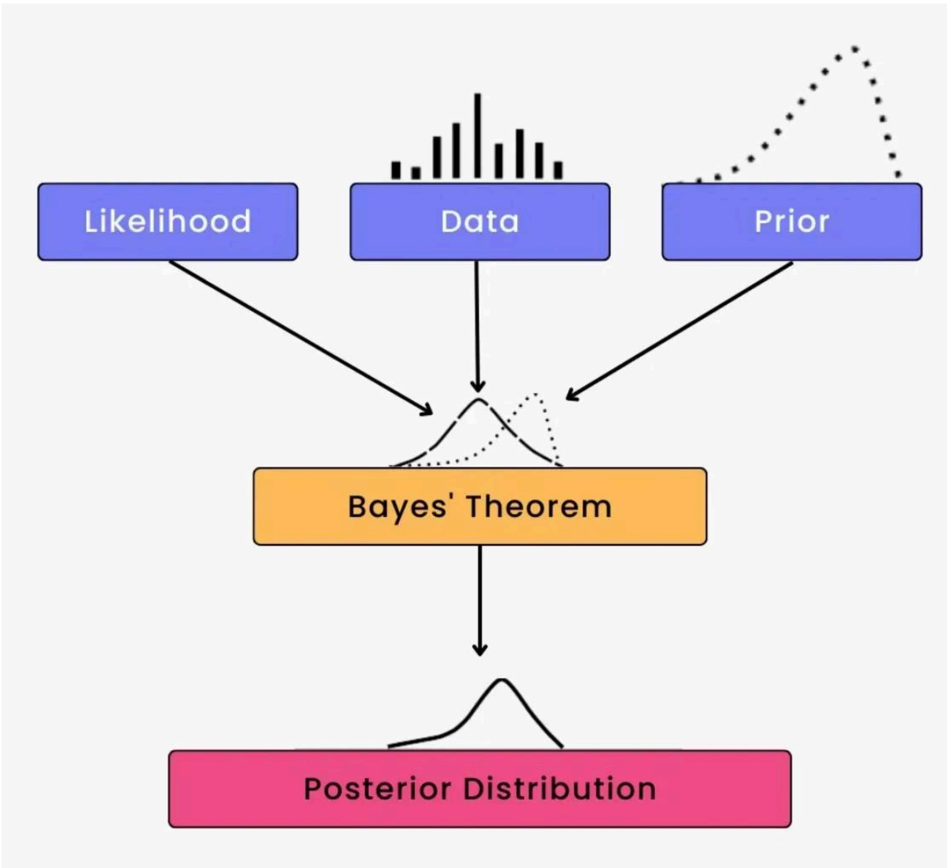
Машинное обучение часто представляют как систему, которая получает данные и выдает прогнозы. Однако в реальных инженерных задачах редко можно быть полностью уверенным. Датчики дают погрешность, измерения могут быть неполными, условия окружающей среды меняются, и не всегда можно точно предсказать будущие события.

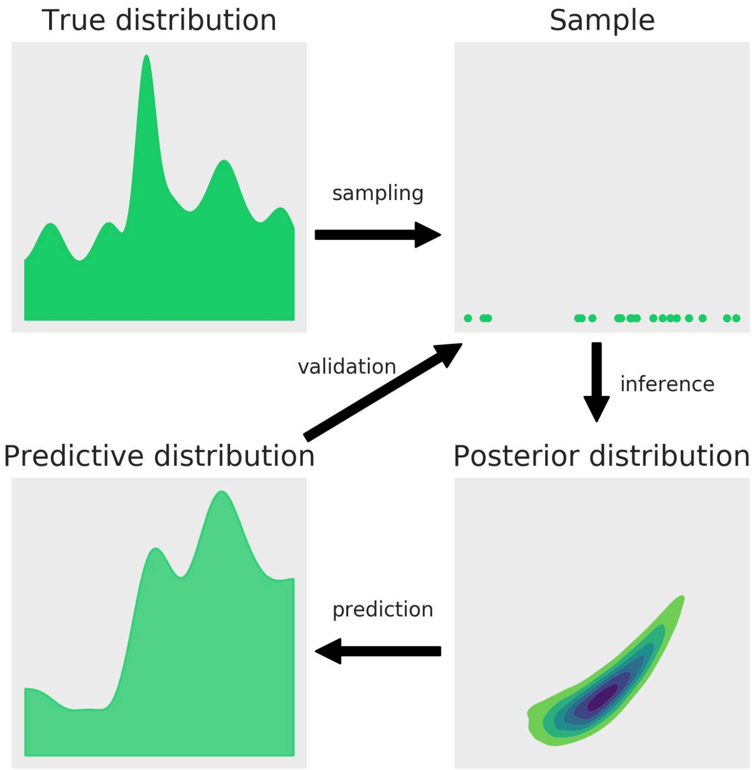
Вероятностное машинное обучение (ВМО) устраняет эту неопределенность, представляя прогнозы и взаимосвязи с помощью вероятностных распределений, а не полагаясь только на однозначные числовые ответы. Вместо вопроса *«Что произойдет?»* вероятностная модель может задать вопрос:

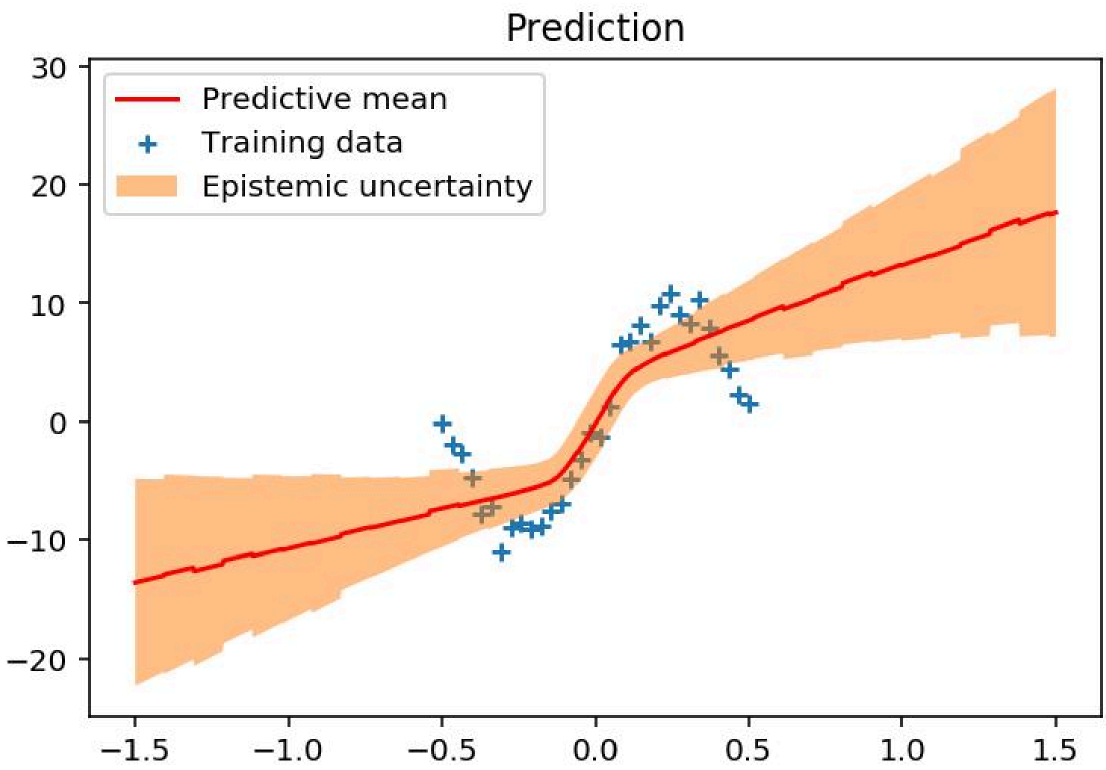
> ***«Что может произойти, насколько вероятна каждая из возможных ситуаций и насколько мы в этом уверены?»*** 🎯

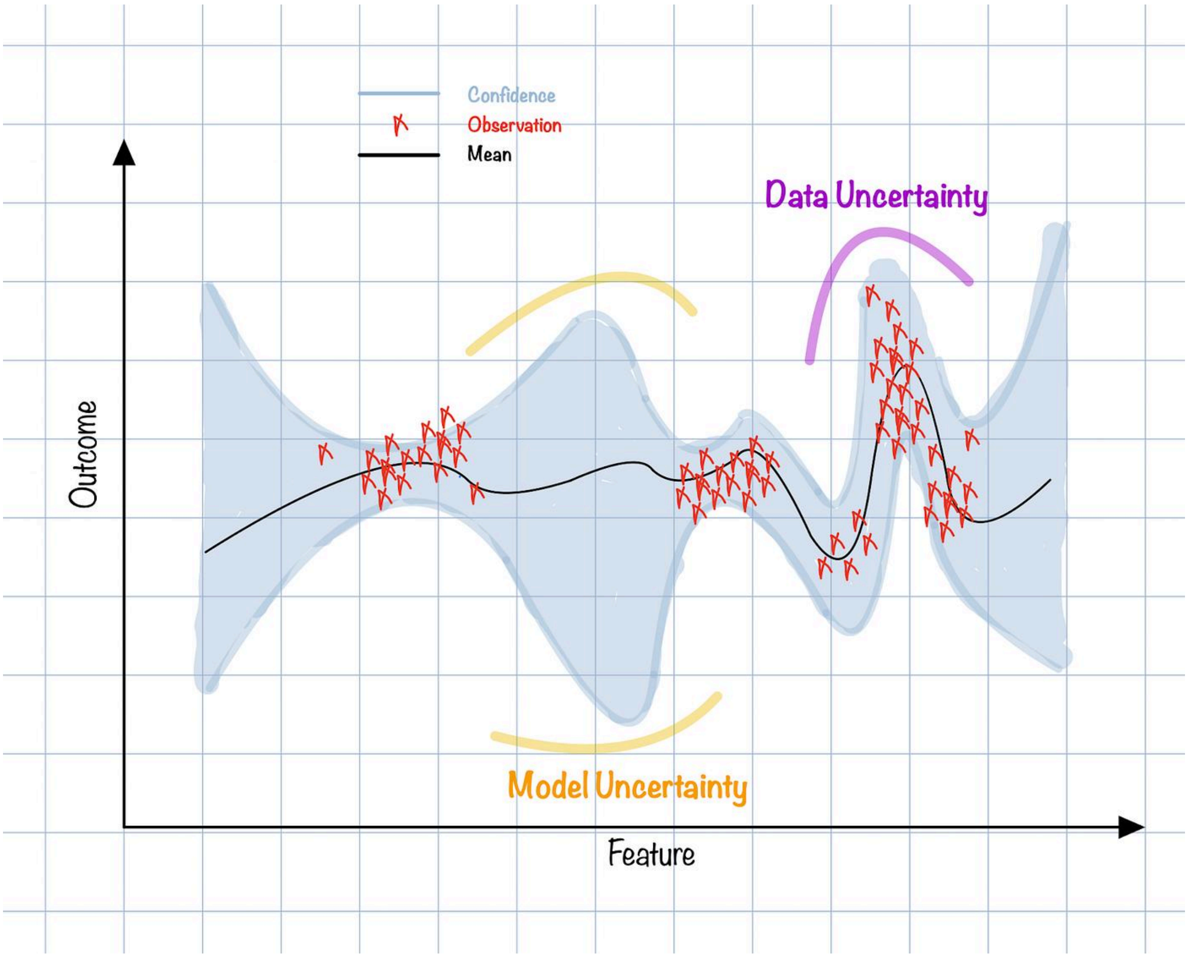
Такой подход особенно ценен в инженерном деле, поскольку неопределенность неизбежна. Датчик температуры может показывать 75 °C, но фактическая температура может быть немного выше или ниже. Модель профилактического обслуживания может оценить вероятность выхода из строя подшипника в течение определенного периода в 78 %, а не просто указать «вышел из строя» или «не вышел из строя».

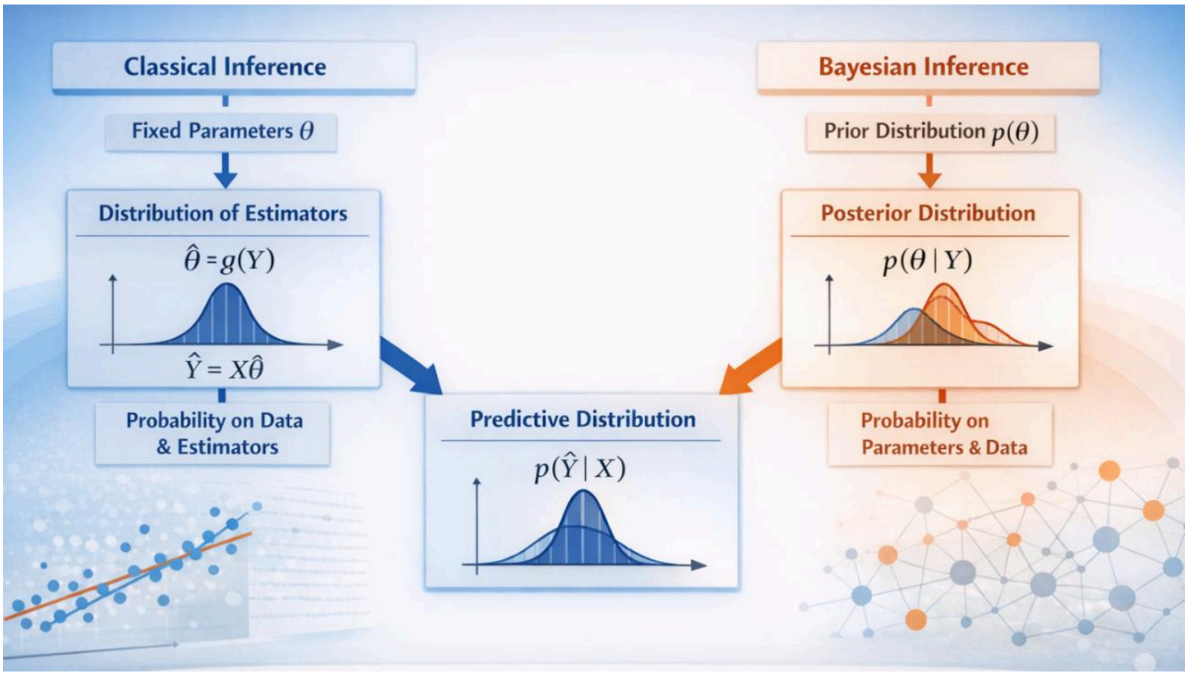





True distribution
Sample
sampling
validation
inference
Predictive distribution
Posterior distribution
prediction


Prediction
Predictive mean
Training data
Epistemic uncertainty
30
20
10
0
−10
−20
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5





Вероятностные методы сочетают в себе математическую теорию вероятности, статистику, оптимизацию и машинное обучение. Их можно применять в робототехнике, аэрокосмической отрасли, автономных транспортных средствах, финансовом инжиниринге, медицинских технологиях, промышленной автоматизации, энергетических системах, телекоммуникациях и многих других областях.

Для студентов PML — это связующее звено между классической статистикой и современным искусственным интеллектом. Для профессиональных инженеров — это систематическая основа для принятия решений в условиях неопределенности данных и знаний.

Теоретическая база

От детерминированных моделей к вероятностным

Детерминированная модель пытается выдать один определенный результат для заданных входных данных:

$$y=f(x)$$

Если x известно, модель предполагает, что y определено.

Вероятностная модель представляет результат в виде распределения.

$$p(y \mid x)$$

Здесь p(y|x) описывает вероятность различных значений y при условии x.

Например, беспилотный автомобиль может оценить:

$$P(\text{пешеходный переход} \mid \text{данные с датчиков})=0{,}93$$

Результат содержит гораздо больше информации, чем простое бинарное предсказание.

Вероятность как инженерный язык

Вероятность предоставляет инструменты для представления неопределенных событий:

$$0 \le P(A) \le 1$$

где P(A) обозначает вероятность события A.

Два фундаментальных понятия:

Априорная вероятность

$$P(\boldsymbol{\theta})$$

отражает наши представления о параметре до получения новых данных.

Вероятность

$$P(D \mid \boldsymbol{\theta})$$

описывает, насколько наблюдаемые данные D согласуются с конкретным параметром θ.

Байесовский вывод объединяет эти понятия:

$$P(\boldsymbol{\theta} \mid D)=D(P)D(P \mid \boldsymbol{\theta})P(\boldsymbol{\theta})$$

где:

-)D|θ(P = апостериорная вероятность
-)θ|D(P = правдоподобие
-)θ(P = априорная вероятность
-)D(P = свидетельства

Это уравнение — одна из основ вероятностного машинного обучения. 🔬

Definition

Probabilistic Machine Learning is a branch of machine learning that uses probability theory and statistical inference to model relationships in data, make predictions, quantify uncertainty, and support decisions under incomplete or noisy information.

Unlike approaches that focus exclusively on minimizing prediction error, probabilistic machine learning attempts to understand the probability distribution underlying observations.

A general probabilistic model can be represented as:

$$p(X,Y,\boldsymbol{\theta})$$

where:

- X represents observed inputs,

- Y represents outputs or targets,
- θ represents model parameters.

Depending on the application, the model may estimate:

$$p(Y \mid X)$$

or infer hidden variables:

$$p(Z \mid X)$$

where Z represents latent or unobserved information.

This makes PML especially useful when an engineering system must distinguish between **prediction** and **confidence in prediction**.

Step-by-Step Explanation of Probabilistic Machine Learning

Step 1: Define the engineering problem

Begin by identifying the quantity that needs to be predicted or estimated.

For example:

> *Predict whether an industrial motor will experience a fault within the next 24 hours.*

Possible inputs include:

- vibration amplitude,
- motor temperature,
- current consumption,
- operating speed,
- historical maintenance records.

The target variable could be:

$$Y \in \{0,1\}$$

where 1 represents a predicted fault.

Step 2: Collect and prepare data

Data quality strongly influences probabilistic models.

An engineering dataset might look like:

Temperature	Vibration	Current	Fault
61°C	2.1 mm/s	8.2 A	0
68°C	3.5 mm/s	8.9 A	0
77°C	5.2 mm/s	9.8 A	1
82°C	6.4 mm/s	10.4 A	1

Measurements should be checked for missing values, sensor errors, outliers, inconsistent units, and sampling problems.

Step 3: Select a probabilistic model

The appropriate model depends on the problem.

Common choices include:

- Bayesian linear regression
- Logistic regression
- Gaussian processes
- Bayesian networks
- Hidden Markov models
- Mixture models
- Probabilistic graphical models
- Probabilistic neural networks
- Bayesian deep learning

For a binary failure prediction problem, logistic regression may be sufficient:

$$P(Y=1 \mid X)=\sigma(w^T X+b)$$

where:

$$\sigma(z)=\frac{1}{1+e^{-z}}$$

Step 4: Estimate parameters

The model parameters must be learned from data.

A maximum-likelihood approach estimates:

$$\hat{\theta} = \arg\max_{\theta} P(D \mid \theta)$$

A Bayesian approach incorporates prior knowledge:

$$P(\theta \mid D) \propto P(D \mid \theta) P(\theta)$$

This is particularly useful when data is limited but engineering knowledge is available.

Step 5: Generate predictions

Instead of returning only a class, a probabilistic model can provide:

$$P(\text{failure}) = 0.82$$

This means the model estimates an 82% probability under its assumptions and data—not that failure is guaranteed.

Step 6: Quantify uncertainty

One of the biggest advantages of PML is uncertainty estimation.

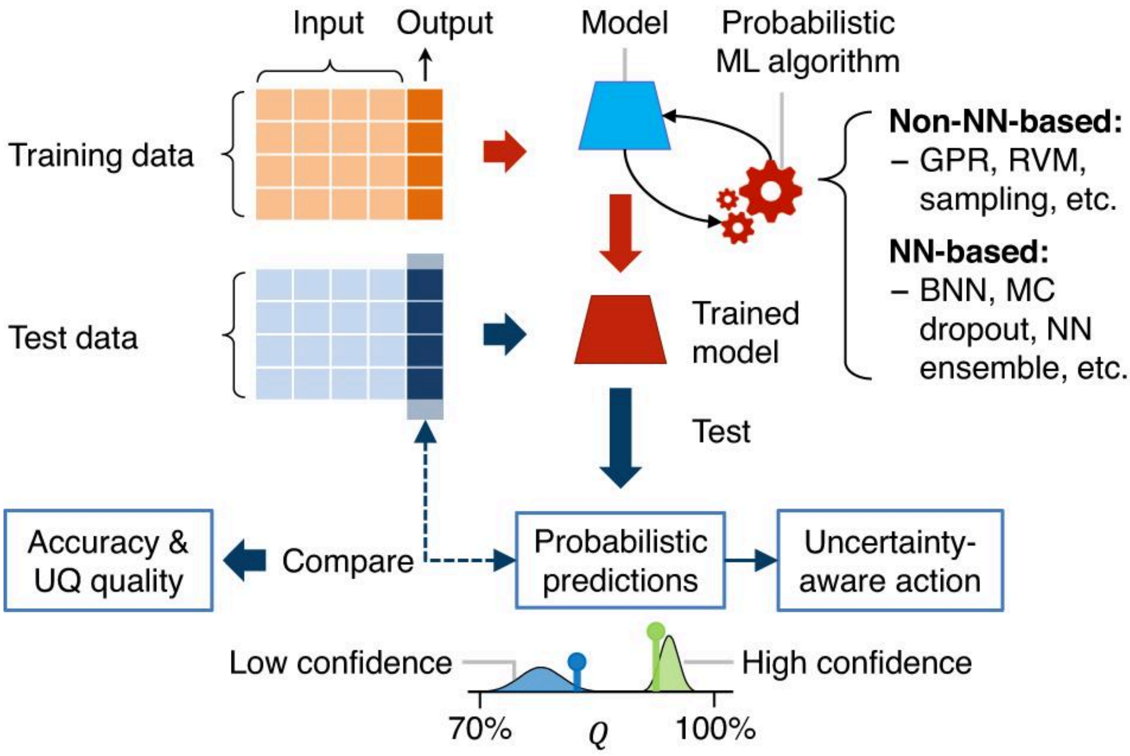
For example:

$$\hat{y} = 72^\circ C$$

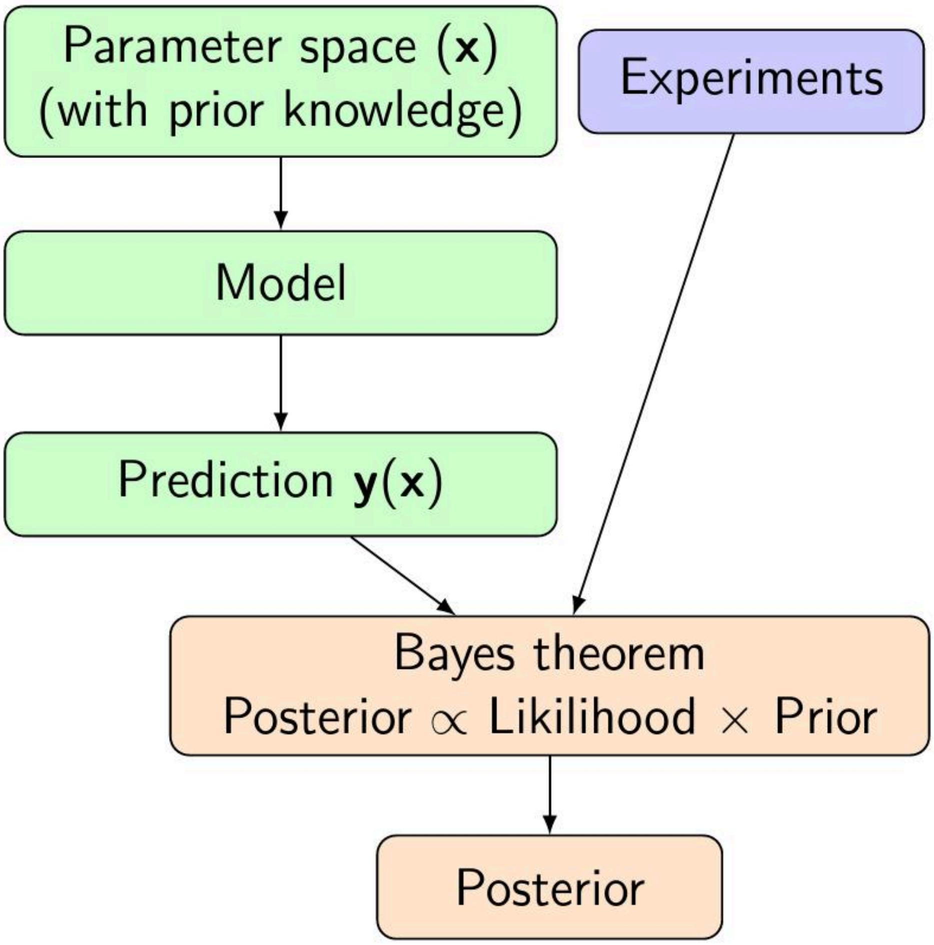
may be accompanied by:

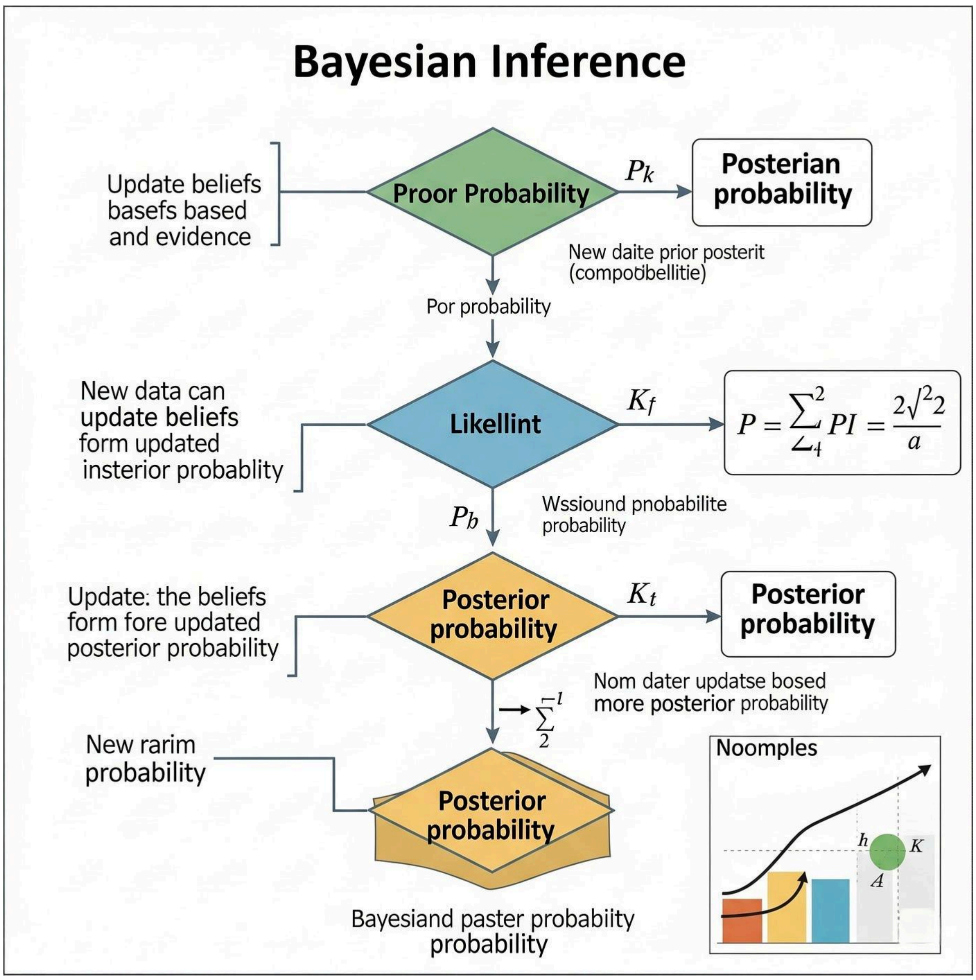
$$72^\circ C \pm 4^\circ C$$

or by a complete predictive distribution.



Input
Output
Model
Probabilistic ML algorithm
Training data
Non-NN-based:
– GPR, RVM, sampling, etc.
NN-based:
– BNN, MC dropout, NN ensemble, etc.
Test data
Trained model
Test
Accuracy & UQ quality
Compare
Probabilistic predictions
Uncertainty-aware action
Low confidence
High confidence
70%
Q
100%


Parameter space (x)
(with prior knowledge)
Experiments
Model
Prediction y(x)
Bayes theorem
Posterior ∝ Likilihood × Prior
Posterior


Bayesian Inference
Update beliefs
basefs based
and evidence
Proor Probability
P_k
Posterian probability
New daite prior posterit
(compotibellitie)
Por probability
New data can
update beliefs
form updated
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K_f
P_b
Wssiound pnobabilite
probability
Update: the beliefs
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posterior probability
Posterior probability
K_t
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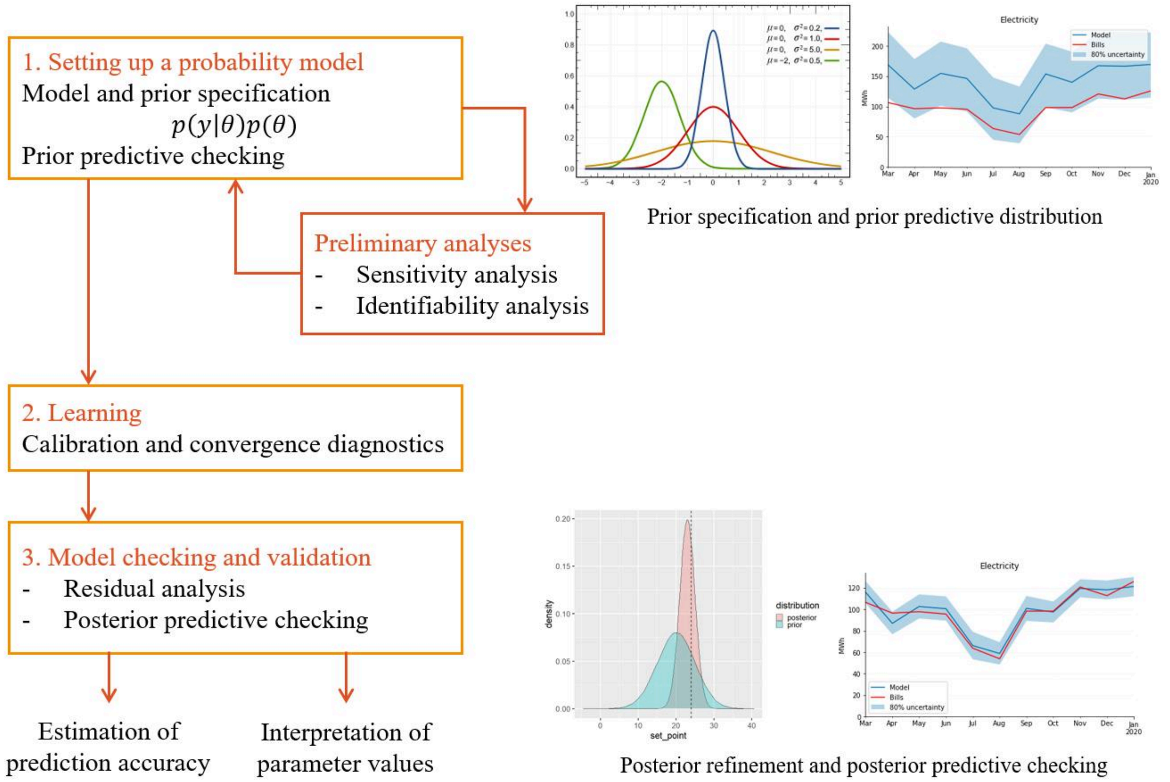

Step 7: Validate the model

A good probabilistic model should not only predict accurately; its probabilities should also be meaningful.

Important evaluation measures include:

- log loss,
- Brier score,
- calibration error,
- likelihood,
- predictive accuracy,
- precision and recall,
- uncertainty quality.

A model predicting 90% probability should ideally be correct approximately 90% of the time among similarly confident predictions.

Comparison

Probabilistic vs deterministic machine learning

Feature	Deterministic approach	Probabilistic approach
Output	Single prediction	Distribution/probability
Uncertainty	Often implicit	Explicit
Noise handling	Model-dependent	Naturally represented
Decision support	Limited confidence information	Confidence-aware
Prior knowledge	Usually less direct	Can be incorporated
Complexity	Often simpler	Can be computationally demanding
Engineering risk analysis	Moderate	Strong
Missing information	Depends on model	Can be explicitly modeled

Neither approach is universally superior.

If an engineer needs a fast temperature estimate for a control loop, a deterministic model may be perfectly adequate. If the system must decide whether an expensive aircraft component requires immediate inspection, uncertainty becomes much more important.

Diagrams and Tables

A simple probabilistic learning architecture

```
┌────────────────────┐
│ Engineering        │
│ Data               │
└─────────┬──────────┘
          ↓
┌────────────────────┐
│ Data Cleaning      │
│ & Features         │

```

```
└──────────┬──────────┘
           ↓
┌─────────────────────┐
│ Probabilistic     │
│ Model             │
└──────────┬──────────┘
           ↓
┌─────────────────────┐
│ Probability       │
│ Distribution      │
└──────────┬──────────┘
           ↓
┌─────────────────────┐
│ Decision +        │
│ Uncertainty       │
└─────────────────────┘
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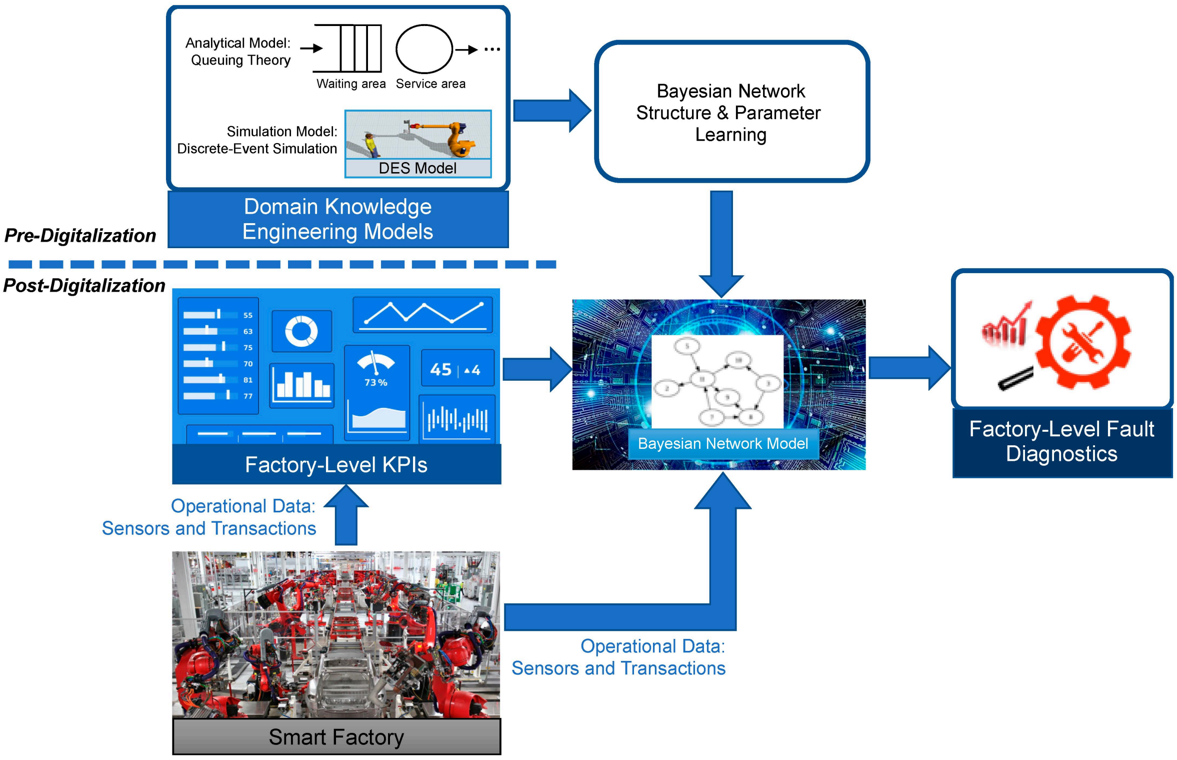
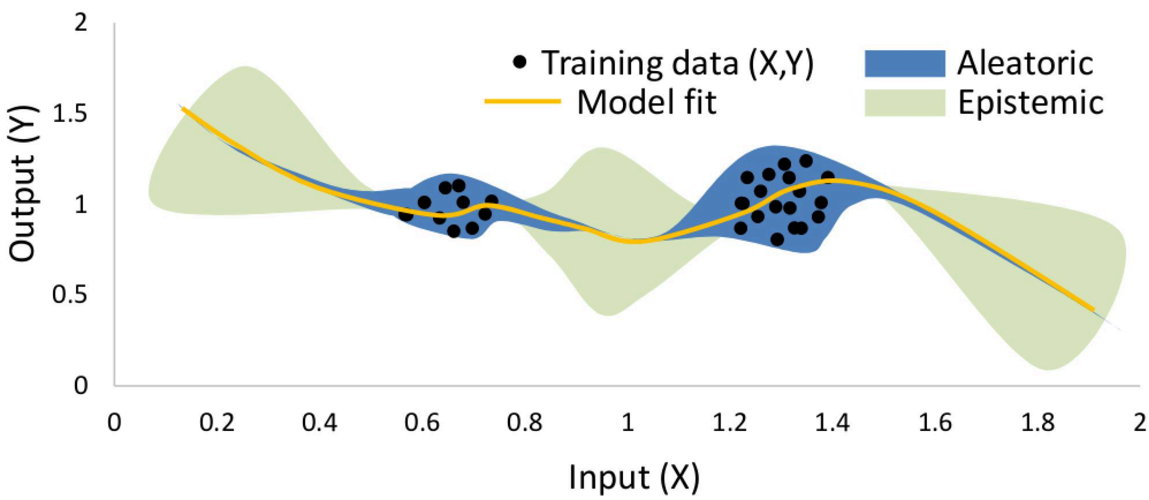
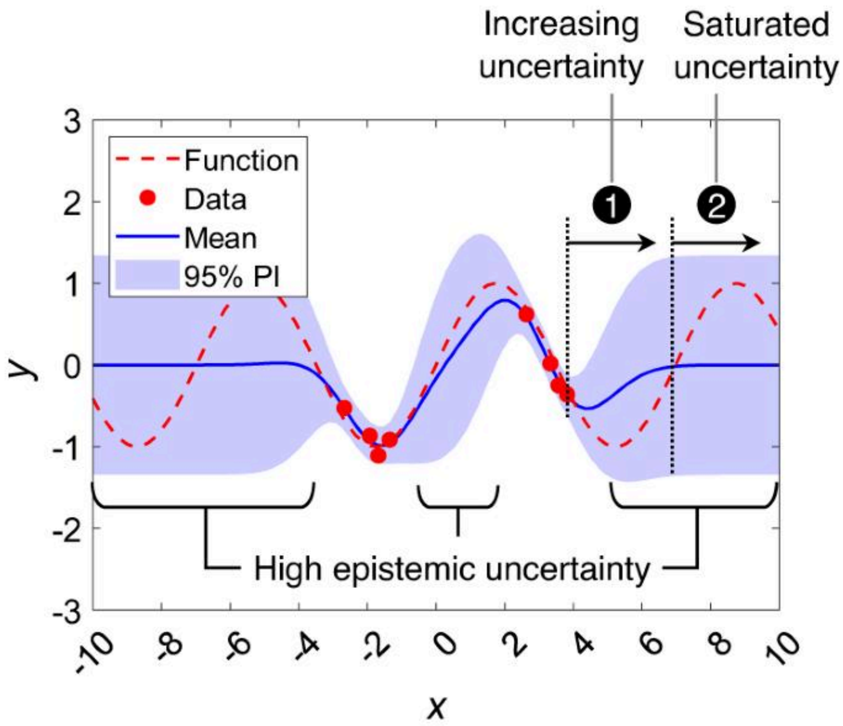
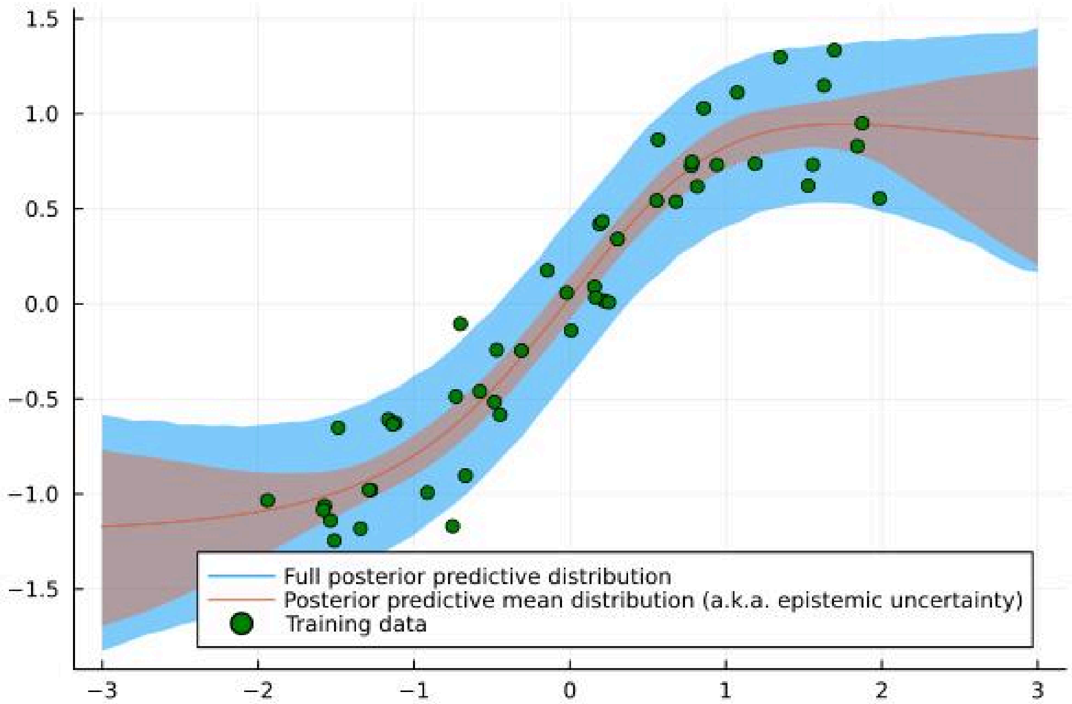


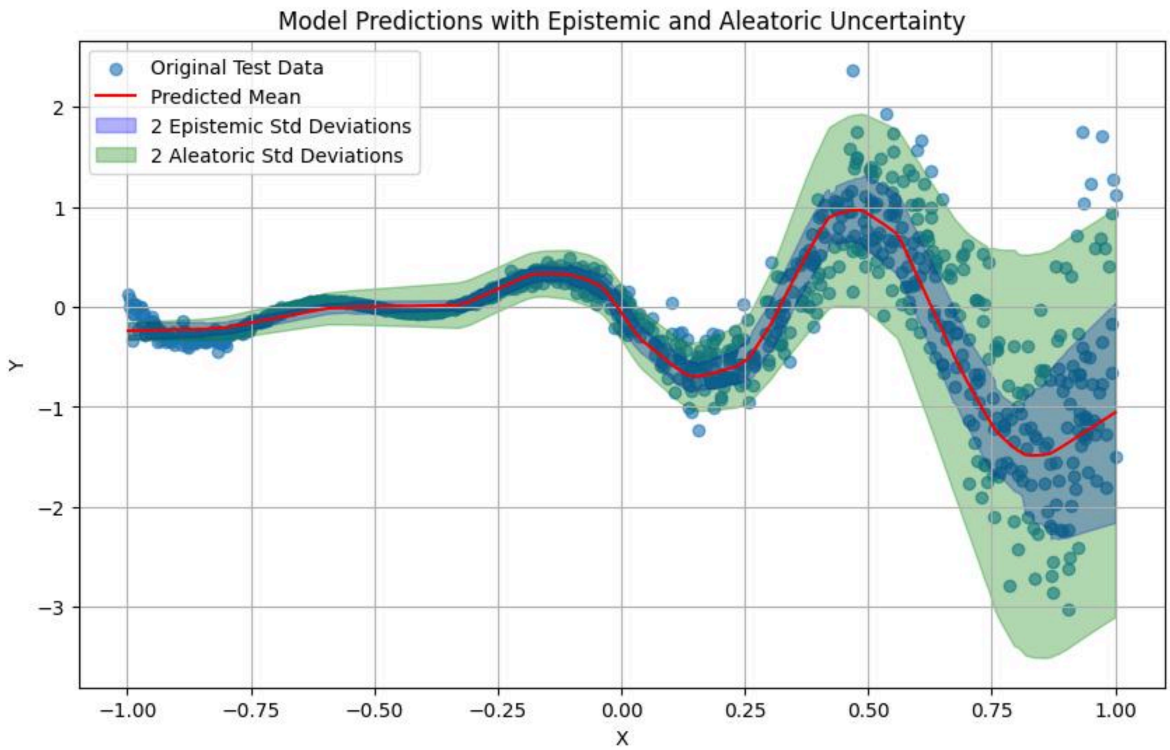




Increasing uncertainty
Saturated uncertainty
Function
Data
Mean
95% PI
High epistemic uncertainty
y
x



Full posterior predictive distribution
Posterior predictive mean distribution (a.k.a. epistemic uncertainty)
Training data



Major probabilistic model families

Model family	Typical use	Engineering example
Bayesian regression	Continuous prediction	Temperature estimation
Logistic regression	Binary outcomes	Fault detection
Gaussian process	Small-data regression	Sensor calibration
Bayesian network	Dependency modeling	Risk analysis
Hidden Markov model	Sequential states	Machine condition monitoring
Mixture model	Clustering	Operating-mode identification
Bayesian neural network	Complex nonlinear prediction	Autonomous systems

Examples

Example 1: Predicting equipment failure

Suppose a pump-monitoring system receives vibration and temperature measurements.

The model estimates:

$$P(F=1 \mid X)=0.87$$

The result indicates an estimated 87% probability of failure according to the trained model.

An engineer might establish a decision threshold:

$$P(F=1 \mid X)>0.80$$

If the threshold is exceeded, the system can recommend inspection.

The important point is that the probability does not automatically equal a physical certainty. The model's calibration, training data, sensor quality, and assumptions must all be considered.

Example 2: Robot localization

A robot rarely knows its exact position.

Instead, it may represent position using:

$$P(x,y)$$

If a robot believes it is most likely at coordinates (4,7), uncertainty may spread around that location.

As new sensor measurements arrive, the distribution changes.

This is the basic idea behind probabilistic localization techniques such as Bayesian filtering.

Example 3: Demand forecasting

An energy company may need to predict electricity demand.

Instead of forecasting:

$$D=500 \text{ MW}$$

a probabilistic model might estimate:

$$P(D \mid X)$$

with a high-probability interval around the forecast.

This allows engineers and operators to plan reserve capacity more intelligently.

Real-World Applications

Predictive maintenance

Industrial facilities use probabilistic models to estimate equipment degradation and failure risk.

Potential inputs include:

- vibration,
- acoustic signals,
- temperature,
- pressure,
- electrical current,
- maintenance history.

The output can support maintenance scheduling before catastrophic failure occurs. ⚙️

Autonomous vehicles

Autonomous systems must reason about uncertain environments.

A vehicle may estimate the probability that an observed object is:

$$P(\text{vehicle}), P(\text{pedestrian}), P(\text{cyclist})$$

These probabilities can contribute to safer planning and control.

Aerospace engineering

Aircraft systems operate under uncertain environmental and mechanical conditions.

Probabilistic methods can support:

- reliability analysis,

- fault diagnosis,
- trajectory estimation,
- sensor fusion,
- structural risk assessment.

Energy systems

Power grids experience uncertainty from demand fluctuations, renewable generation, equipment failures, and weather.

Probabilistic forecasting can help estimate future demand and generation ranges rather than relying exclusively on point forecasts.

Telecommunications

Wireless systems face uncertain channel conditions, interference, traffic patterns, and user behavior.

Probabilistic models can assist with:

- channel estimation,
- network optimization,
- traffic prediction,
- anomaly detection,
- resource allocation.

Common Mistakes

Treating probability as certainty

An 80% probability does **not** mean an event must occur.

Probability represents uncertainty under a defined model and dataset.

Ignoring calibration

A classifier may achieve high accuracy while producing poorly calibrated probabilities.

For risk-sensitive engineering, calibration deserves separate attention.

Using an unnecessarily complicated model

A sophisticated Bayesian neural network is not automatically better than a simple probabilistic regression model.

Model complexity should be justified by the engineering problem.

Ignoring data quality

A mathematically elegant model cannot compensate for unreliable sensors or biased training data.

Confusing correlation with causation

Probabilistic relationships do not automatically prove that one variable causes another.

Forgetting uncertainty in the data

Measurement uncertainty should be considered alongside model uncertainty whenever possible.

Challenges and Solutions

Challenge	Possible solution
Limited training data	Use informative priors or domain knowledge
Noisy sensors	Explicitly model measurement noise
High computational cost	Approximate inference or simpler models
Poor probability calibration	Calibration techniques and validation
Complex posterior distributions	Sampling or variational inference
Model overconfidence	Regularization and uncertainty evaluation
Distribution shift	Monitor incoming data continuously

Computational complexity

Exact Bayesian inference can become difficult as models grow.

For example, calculating:

$$P(\theta \mid D)$$

may require integrating over a large parameter space.

Modern methods therefore use techniques such as:

- Markov Chain Monte Carlo,
- variational inference,
- Laplace approximations,
- sequential Monte Carlo,
- approximate Bayesian inference.

The engineering objective is usually not to find the most mathematically complicated method, but to find a method that provides an appropriate balance between **accuracy, uncertainty, computational cost, and reliability**.

Case Study: Probabilistic Predictive Maintenance

Consider a manufacturing facility operating hundreds of electric motors.

Historically, maintenance was performed according to fixed schedules. This creates two problems:

1. Components may be replaced before they actually need replacement.
2. Components can fail unexpectedly between scheduled inspections.

The engineering team introduces a probabilistic predictive-maintenance system.

Data collection

Sensors continuously measure:

$$X=[T,V,I,S]$$

where:

- T = temperature,
- V = vibration,
- I = current,

- S = rotational speed.

Model

A probabilistic classifier estimates:

$$P(F=1 \mid X)$$

for each motor.

Suppose Motor A produces:

$$P(F=1 \mid X)=0.12$$

while Motor B produces:

$$P(F=1 \mid X)=0.91$$

Motor B receives a higher maintenance priority.

Engineering decision

The maintenance system could define three risk zones:

Probability	Risk level	Suggested action
0–0.30	Low	Normal monitoring
0.30–0.70	Medium	Increased inspection
0.70–1.00	High	Immediate evaluation

The thresholds should be selected using actual failure costs, operational requirements, and validation data—not arbitrarily.

This approach transforms machine learning from a simple prediction tool into a **risk-aware engineering decision system**. 🚀

Essential Tips

Start with probability fundamentals

Students should become comfortable with:

$$P(A), P(A \mid B), E[X], Var(X)$$

before jumping into advanced probabilistic deep learning.

Understand distributions

Important distributions include:

- Gaussian,
- Bernoulli,
- Binomial,
- Poisson,
- Exponential,
- Beta,
- Gamma.

Understanding when each distribution is appropriate is often more valuable than memorizing formulas.

Visualize uncertainty

Do not look only at a predicted value.

Plot:

- probability distributions,
- confidence or credible intervals,
- predictive intervals,
- residual distributions,
- calibration curves.

Combine domain knowledge with data

Engineering knowledge can provide valuable prior information.

A model that respects physical constraints can sometimes outperform a purely data-driven model, especially when training data is limited.

Validate probabilistic predictions

Ask two separate questions:

Is the model accurate?

and

Are its probabilities trustworthy?

Both matter.

FAQs

What is probabilistic machine learning?

Probabilistic machine learning uses probability and statistical inference to model data, make predictions, and quantify uncertainty.

How is probabilistic machine learning different from traditional machine learning?

Traditional machine learning often focuses on predicting an output, while probabilistic machine learning explicitly represents uncertainty and probability distributions.

Is Bayesian machine learning the same as probabilistic machine learning?

Not exactly. Bayesian machine learning is an important part of probabilistic machine learning, but the broader field includes many probabilistic approaches that are not necessarily fully Bayesian.

What mathematics is required to learn PML?

A useful foundation includes probability, statistics, linear algebra, calculus, and optimization. Engineers do not need to master every mathematical topic before beginning practical applications.

Is probabilistic machine learning useful for engineers?

Yes. It is especially useful for systems involving noisy sensors, uncertain measurements, reliability, forecasting, diagnosis, robotics, control, and risk analysis.

What is Bayesian inference?

Bayesian inference updates beliefs about unknown parameters or hypotheses using observed evidence:

$P(\theta|D) \propto P(D|\theta)P(\theta)$

It provides a formal mechanism for combining prior information with new data.

Does probabilistic machine learning always produce better predictions?

No. Its major advantage is not automatically higher predictive accuracy. Its strength is providing a structured representation of uncertainty and supporting decisions when predictions are imperfect.

What should beginners learn first?

Start with probability, statistics, distributions, conditional probability, Bayes' theorem, regression, and basic machine learning. Then progress to Bayesian inference, graphical models, Gaussian processes, and probabilistic deep learning.

Conclusion

Probabilistic Machine Learning provides an engineering-oriented way to reason about uncertainty. Instead of treating every prediction as an unquestionable number, it represents the range of possible outcomes and the probability associated with them.

The fundamental idea can be summarized as:

$\text{Data} + \text{Probability} + \text{Inference} \rightarrow \text{Uncertainty-Aware Decisions}$

This is particularly important for modern engineering systems. Robots must estimate uncertain positions, industrial machines must operate despite noisy sensors, energy systems must handle uncertain demand, and autonomous vehicles must make decisions in unpredictable environments.

For beginners, probabilistic machine learning offers an excellent way to connect statistics with artificial intelligence. For experienced engineers, it provides practical tools for reliability, prediction, diagnosis, optimization, and risk management.

The most important lesson is simple: **a good machine-learning system should not only tell engineers what it predicts—it should also communicate how uncertain that prediction is.** 🎯🤖

As engineering systems become more autonomous and data-driven, that distinction will become increasingly important across the USA, UK, Canada, Australia, and Europe.

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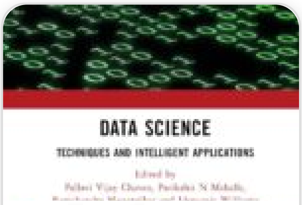

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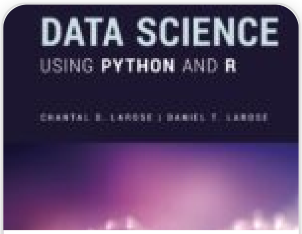

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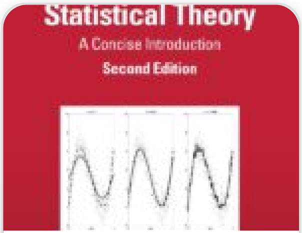

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